Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every ~7 Months

50% time horizon doubles every ~207 days ($R^2 = 0.97$)

Kwa, West, Becker, et al. — METR | arXiv:2503.14499 | March 2025

A New Metric: 50% Task-Completion Time Horizon

1

Assemble Task Suite

170 diverse software & research tasks spanning 1 second to 8 hours of skilled human effort

2

Measure Humans & AI

Time skilled professionals on each task; run AI agents and record success rates

3

Fit Logistic Model

Estimate the task duration at which each AI achieves exactly 50% success probability

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Figure 2. Methodology overview

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

HCAST
97 tasks

Diverse software tasks | 1 min – 30 hrs

RE-Bench
7 tasks

Difficult ML research engineering | ~8 hrs each

SWAA
66 tasks

Short single-step actions | 1 sec – 30 sec

Task Family	Human Time	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?"
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 minutes	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30x speedup of a backtesting tool

Over 800 human baselines totaling 2,529 hours collected across all tasks

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes (o3)

~207 days
Doubling Time | ≈ 7 months

$R^2 = 0.97$
Fit Quality | 6-year trend

~2 sec
2019 Baseline | GPT-2

~1h 50m
2025 Frontier | o3

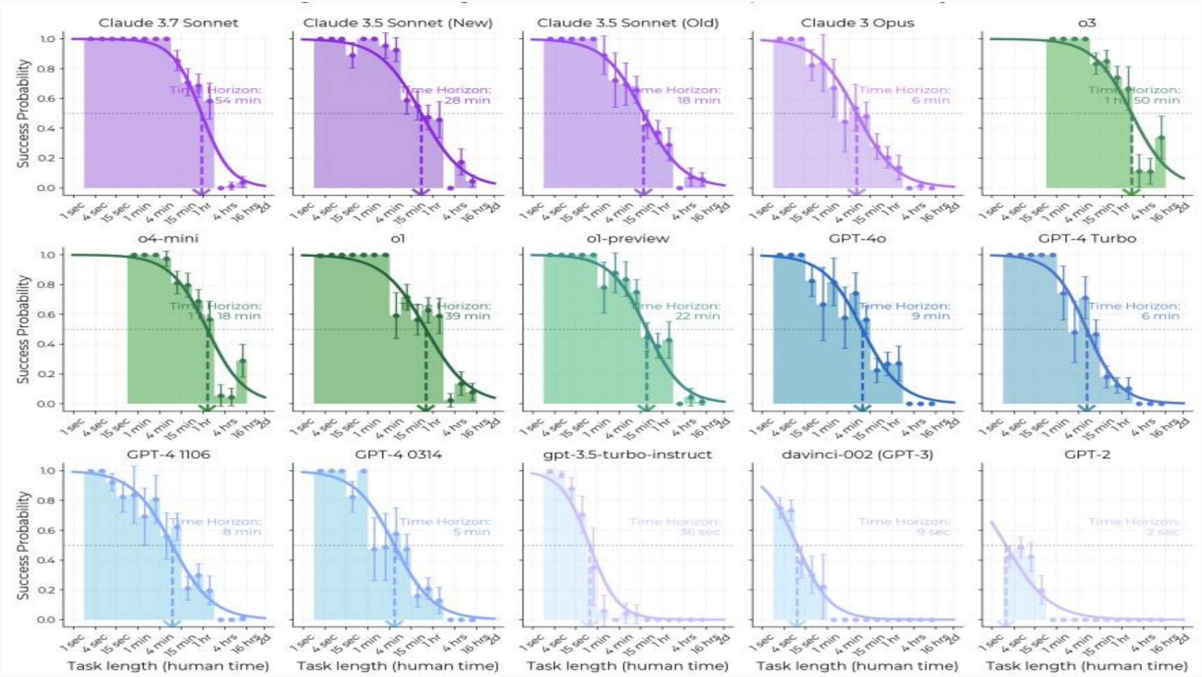


Figure 1. Exponential trend in AI time horizon (50% success) vs. model release date

Success Probability Drops Sharply as Task Length Increases for All Models

Frontier models push the curve rightward

- GPT-2 (2019): ~2 sec horizon
- GPT-4 (2023): ~5 min horizon
- Claude 3.5 Sonnet (2024): ~28 min
- o1 (2024): ~39 min
- Claude 3.7 Sonnet (2025): ~54 min
- o3 (2025): ~1 hr 50 min

The 80% time horizon shows a similar trend but with horizons roughly 5× shorter.

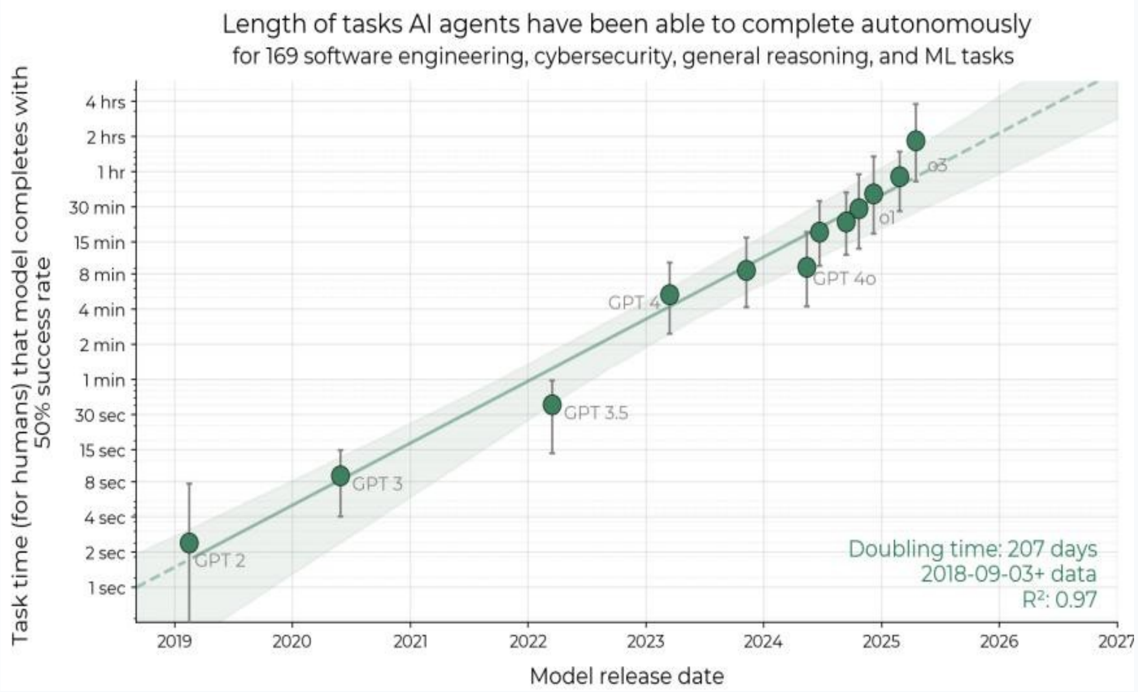


Figure 3. Per-model success probability curves

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Improved Logical Reasoning

Models can follow longer chains of inference required for multi-step software tasks

Better Tool Use

More effective integration with IDEs, terminals, debuggers, and external APIs

Greater Reliability & Self-Awareness

Models recognize when they are stuck and can request clarification or backtrack

Ability to Adapt to Mistakes

Error recovery and iterative debugging rather than failing on first misstep

Limitation

Performance is notably lower on less structured, "messier" tasks. External validity remains an open question — the benchmark tasks may not perfectly represent real-world intellectual labor distributions.

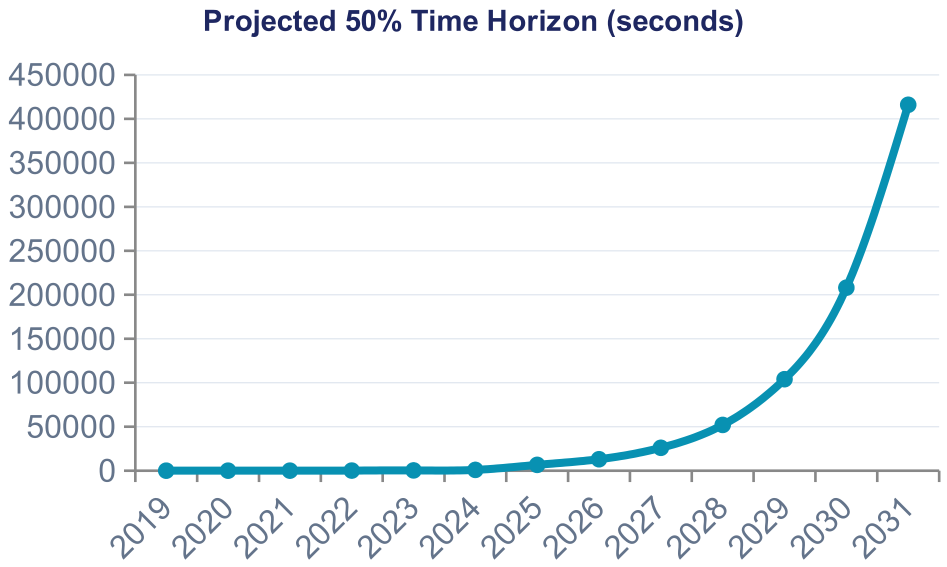
Extrapolation Predicts Month-Long Task Automation by 2028–2031

1 Month

(167 work hours)

Estimated arrival: mid-2028 to mid-2031 under naive extrapolation of the 2019–2025 trend.

2023–2025 growth rate is ~20% faster than the overall 2019–2025 rate, suggesting possible acceleration.



Naive extrapolation — actual trajectory may differ due to trend changes or capability plateaus

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1. 50% time horizon is a robust, intuitive metric for tracking AI autonomy with $R^2 = 0.97$ over 6 years of model releases.
2. Doubling every ~7 months with possible acceleration since 2024 (2023–2025 rate ~20% faster).
3. Current frontier: o3 completes tasks requiring ~2 hours of skilled human-equivalent work autonomously.
4. External validity remains an open question — tasks may not perfectly represent real-world intellectual labor distributions.
5. Key limitations: poor performance on unstructured “messier” tasks and uncertain generalization to real-world job distributions.